

Real-time analytic sensitivity method for transient security assessment and preventive control

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Abstract: An analytic sensitivity method is proposed for the transient security assessment of large power systems. It is based on the recently developed 'extended equal area criterion', and on a related transient stability margin. It is shown that this margin leads to a simple algebraic formulation of sensitivity coefficients. Particular attention is focused on the sensitivity with respect to generation power, and possible applications to power system planning and online operation are suggested. For example, the computation of generation and load supply limits provides transient stability constraints and thence means of online transient security assessment and preventive control. Within the limits of the simplified electromotive force behind the transient reactance model, the method proves to be accurate, demanding very little with respect to computing time and fairly independent of the system size. The simulation results reported in the paper corroborate these conclusions.

1 Introduction

In principle, numerical integration methods can be used to assess transient stability of power systems with any degree of modelling sophistication. However, the dimensionality burden increases significantly with the system size. This makes the numerical integration methods rather awkward for real-world planning screening applications, where thousands of cases have to be run, or for online operation where computation requirements become very stringent. For this kind of uses, the Liapunov direct criterion along with recent practical improvements proves to be an interesting alternative, provided that simplified system modelling is deemed acceptable [1, 2]. Another direct method, the so-called *extended equal area criterion* (EEAC) was recently proposed [3, 4]. This one-shot procedure has proved to be substantially faster than the Liapunov criterion. It also opens real possibilities towards analytic sensitivity analysis.

Sensitivity analysis in general is an extremely useful, arguably vital, function of power system security assessment and control. In the early stage, when only steady-

state security assessment and eigenvalue analysis of power systems were developed, derivatives were used to find search directions towards an optimum or simply acceptable solution. More recently, strong interest in transient sensitivity methods was also felt. Indeed, what essentially matters in practice is to assess stability limits (power generation and transmission on important lines, or load supply capabilities). And what essentially interest the system operator are guidelines enabling him to face the 'what if' question, whenever it arises. Besides, transient sensitivity analysis is also the basis of transient security dispatch, preventive control, system reinforcement and determination of the needed degree of modelling sophistication.

Likewise steady-state, transient sensitivity analysis may be approached through either repeated simulations or analytical formalism. The former results in excessively long and expensive runs and possibly in too large truncation or conditioning errors, whenever the perturbation step size is inappropriate; along these lines, let us mention that the Liapunov direct criterion used repeatedly has been proposed to this purpose [5, 6]. In contrast, analytic sensitivity analysis has definite potential advantages over repeated runs. It can provide sound information about the way the various parameters influence a system's transient behaviour and therefore suggest means of preventive actions. Moreover, it is capable of meeting computational stringent conditions for planning screening and online operation purposes. Yet, as far as we know, no analytic transient sensitivity method is available nowadays.

This paper attempts to lay the foundations of such an approach, on the basis of the EEAC and a related transient stability margin, appropriately chosen. Many results and practical applications can be foreseen. To mention only one, the sensitivity coefficients with respect to generation power take on simple algebraic forms in terms of the system parameters. These in turn allow one to compute transient stability injection limits, and thence to define a transient security region for a given set of selected contingencies. Notice incidentally that even for steady-state security assessment, constructing a security region is far from trivial. Notice also that the transient security region proposed in this paper is not affected by any conservatism, in the sense that it is built on necessary and sufficient conditions (see the discussion of Section 2.4).

2 Extended equal area criterion (EEAC)

The objectives of this section are

- (a) to make the paper self-sufficient
- (b) to introduce the main symbols

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(c) to outline aspects directly related to the subject matter of the paper

(d) to indicate some important differences with the conceptually closely related Liapunov criterion applied to multimachine systems.

More information about the EEAC and related formulas are reported in Appendix 9.

2.1 Fundamentals of the EEAC

2.1.1 Principle: For an assumed disturbance, the multimachine system is decomposed into two subsets. One consists of the cluster of critical machine(s) identified according to the rule set in Reference 4 and recalled in Appendix 9.3; we will henceforth refer to this as the 'critical cluster'. The other comprises the remaining machines. The two subsets are further transformed into two equivalent machines. This amounts to replacing the time evolution of all rotor angles within the same subset by that of its corresponding *partial center of angles* (PCOA). Finally, the equivalent two-machine system is reduced to a 'one-machine-infinite-busbar' system. For this latter, a stability margin corresponding to a given clearing time is defined.

2.1.2 The multimachine system: modelling and assumptions: With the standard simplifying assumptions, the motion of the i th machine of an n -machine system can be described by

$$\begin{aligned} \delta_i &= \omega_i \\ M_i \dot{\omega}_i &= P_{mi} - P_{ei} \quad i = 1, 2, \dots, n \end{aligned} \quad (1)$$

where

$$P_{ei} = E_i^2 Y_{ii} \cos \theta_{ii} + \sum_{j=1, j \neq i}^n E_i E_j Y_{ij} \cos (\delta_i - \delta_j - \theta_{ij}) \quad (2)$$

and where the following standard notation is used:

- δ_i = rotor angle
- ω_i = rotor speed
- M_i = inertia coefficient
- P_{mi} = mechanical input power
- P_{ei} = electrical output power
- E_i = voltage behind the direct axis transient reactance
- Y = admittance matrix reduced at the internal generator nodes
- Y_{ij} = modulus of the ij th element of Y
- θ_{ij} = argument of the ij th element of Y .

M_i , P_{mi} and E_i are assumed to be constant throughout the transients; all loads are modelled as constant impedances.

2.1.3 Equivalent two-machine aggregated model:

Denoting by

s = the specific critical cluster assumed momentarily to consist of one machine only

A = the set of all remaining machines

a = its equivalent, aggregated to one machine

we set [3, 4, 7]

$$\begin{aligned} M_a &= \sum_{l \in A} M_l; \quad \delta_a = M_a^{-1} \sum_{l \in A} M_l \delta_l; \\ \omega_a &= \dot{\delta}_a; \quad \gamma_a = \dot{\delta}_a. \end{aligned} \quad (3)$$

Upon fixing [3, 4]

$$\delta_l = \delta_a \quad \forall l \in A \quad (4)$$

we get the motion of the PCOA of the set A as expressed by

$$M_a \dot{\delta}_a = \sum_{l \in A} (P_{ml} - P_{el}) \quad (5)$$

where

$$\begin{aligned} P_{el} &= E_l^2 Y_{ll} \cos \theta_{ll} + E_l E_s Y_{ls} \cos (\delta_a - \delta_s - \theta_{ls}) \\ &+ \sum_{j \in A, j \neq l} E_l E_j Y_{lj} \cos \theta_{lj} \end{aligned} \quad (6)$$

For the specific critical machine, we get

$$M_s \dot{\delta}_s = P_{ms} - P_{es} \quad (7)$$

with

$$P_{es} = E_s^2 Y_{ss} \cos \theta_{ss} + \sum_{l \in A} E_s E_l Y_{sl} \cos (\delta_s - \delta_a - \theta_{sl}). \quad (8)$$

When the critical cluster comprises more than one machine, we aggregate them to one equivalent, according to the above PCOA rules (eqns. 3 to 6).

2.1.4 The equivalent 'one-machine-infinite-busbar' (OMIB) system: Setting

$$\delta \triangleq \delta_s - \delta_a \quad (9)$$

and using the symbols and parameters defined in Appendix 9.1, we get [4]

$$M \dot{\delta} = P_m - [P_c + P_{max} \sin (\delta - v)]. \quad (10)$$

2.1.5 The extended equal area criterion (EEAC): The EEAC derives from the well known equal area criterion [8], applied to the equivalent multimachine equation of motion (eqn. 10). Fig. 1 illustrates the plot of P/δ curves provided in the pre-fault or original (O), during-fault (D), and post-fault (P) configurations. The original (steady-state) operation is characterised by the rotor angle δ_0 located at the crossing of the horizontal line $P = P_m$ with the original P_{e0} curve, partially drawn. The post-fault stable (respectively unstable) equilibrium point is determined by the intersection of P_m with P_{ep} ; this provides δ_p (respectively $(\pi - \delta_p + 2v_p)$). Moreover, the value that the angle reaches at the fault clearing time delimits the accelerating and decelerating areas, which measure the corresponding transient energies. Let A_{acc} and A_{dec} , respectively, denote these areas.

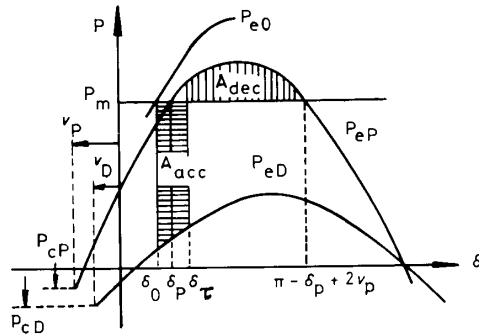


Fig. 1 Principle of the EEAC [4]

O: Original; D: During fault; P: Post-fault

The EEAC simply recognises the fact that the faulted system remains capable of recovering stability as long as A_{acc} is smaller than A_{dec} , and suggests two alternative measures of the system robustness. One is the conventional critical clearing time. Reference 4 has essentially

focused on this measure and proposed algebraic expressions for its one-shot computation.

2.2 Transient stability margin

This alternative measure amounts to assessing 'how far from instability' the system is at the time the considered disturbance is presumably cleared; in the sequel, τ will denote this clearing time. Within the EEAC context, this measure appears to be very interesting; indeed, it opens a way towards analytic sensitivity analysis and thence to control.

The margin is defined as the difference of the decelerating and accelerating areas sketched in Fig. 1:

$$\eta = A_{dec} - A_{acc} \quad (11)$$

i.e. (see eqns. 31 and 32 of Appendix 9.1):

$$\begin{aligned} \eta = & P_m(\delta_0 + \delta_p - \pi - 2v_p) \\ & + P_{cp}(\pi - \delta_p - \delta_\tau + 2v_p) + P_{cd}(\delta_\tau - \delta_0) \\ & + P_{maxp}[\cos(\delta_\tau - v_p) + \cos(\delta_p - v_p)] \\ & - P_{maxd}[\cos(\delta_\tau - v_D) - \cos(\delta_0 - v_D)] \end{aligned} \quad (12)$$

Note that this is an analytic function of the system parameters and of the clearing angle δ_τ . δ_τ in turn may be analytically related to τ via a Taylor series expansion about δ_0 . Denoting the second and fourth derivatives of δ (considered at $t = 0^+$, i.e. immediately after a disturbance inception) by γ and $\ddot{\gamma}$, respectively, and denoting two corrective factors introduced to counterbalance truncation errors by α_1, α_2 , we write (see Appendix 9.1):

$$\delta_\tau = \delta_0 + \alpha_1^{-1} \left[\frac{1}{2} \gamma (\alpha_2^{-1} \tau)^2 + \frac{1}{24} \ddot{\gamma} (\alpha_2^{-1} \tau)^4 \right]. \quad (13)$$

Note that the following expressions could also be proposed as transient stability margins:

$$\eta' = (A_{dec} - A_{acc})/A_{acc}; \quad \eta'' = (A_{dec} - A_{acc})/A_{dec}.$$

Despite their nice dimensionless property, however, these lead to serious subsequent developments. Nevertheless, they could be useful for monitoring purposes.

2.3 Important features and calculation aspects of the EEAC

(a) A prerequisite to the analytic sensitivity analysis is the possibility of expressing analytically the stable equilibrium points of the system in its pre-fault and post-fault configurations. Within the EEAC context these expressions are indeed readily and simply obtained; we find

$$\begin{aligned} \delta_0 &= \sin^{-1} [(P_m - P_{c0})/P_{max0}] + v_0 \\ \delta_p &= \sin^{-1} [(P_m - P_{cp})/P_{maxp}] + v_p \end{aligned} \quad (14)$$

As for the post-fault unstable equilibrium point, it is implicitly replaced by

$$\delta_p^u = \pi - \delta_p + 2v_p. \quad (15)$$

(b) As far as we know, the EEAC is the first fully analytical approach to the calculation of both critical clearing times and transient stability margins. It allows a one-shot transient stability assessment, free from any numerical integration.

(c) Moreover, the EEAC can handle successive actions during the fault-on period.

(d) The margin (eqn. 12) contains a good deal of information about topology, transfer conductances, load levels, power flows and about fault location, type, and duration. Part of this information is exploited in the sequel.

2.4 Energy type Liapunov criteria vs. EEAC

The EEAC owes its numerous interesting properties to the combined contribution of the OMIB equivalent system derived in Section 2.1, and of the Taylor series expressed by eqn. 13. Let us pinpoint the essential similarities and differences between EEAC and energy type Liapunov criteria.

(a) In the trivial single- or two-machine case, the OMIB is indeed the actual system modelling. Moreover, the stability conditions of the Liapunov criteria become identical to those of the EEAC, i.e. they become necessary and sufficient conditions; hence, no conservatism affects either of the two approaches. Their numerical results will be identical if the same Taylor series is used in both approaches.

(b) In the multimachine case, to bring the Liapunov approach close to the EEAC one should embed in the former both of the OMIB equivalent and Taylor series techniques; nevertheless, the EEAC still keeps the advantage of being more tractable thanks to the easy physical interpretations offered by Fig. 1. Otherwise, the two methods have significant differences. Among them, the EEAC features mentioned in Section 2.3 are certainly very important. Besides, deriving from the equal area criterion, the EEAC provides necessary and sufficient stability conditions, whereas the Liapunov method provides sufficient conditions only; in other words, the latter is conservative whereas the former is not, and only engineering know-how has succeeded in overcoming this conservative character of the Liapunov method in some particular cases (e.g. when the post-fault configuration is identical to the pre-fault one). The EEAC, by contrast, is not subject to this kind of restriction.

3 Sensitivity analysis

3.1 General methodology

To estimate the effect of a small change of a parameter x_i on the margin η which is a multivariable function, we calculate its partial derivative

$$\begin{aligned} S_{x_i} &= \lim_{\Delta x_i \rightarrow 0} \frac{\Delta \eta(x_1, x_2, \dots, x_k)}{\Delta x_i} \\ &= \frac{d\eta(x_1, x_2, \dots, x_k)}{dx_i} \end{aligned} \quad (16)$$

Particularly interesting is the case where the above sensitivity coefficient takes on an analytic expression. It is then readily possible to compute its value and hence the value of Δx_i necessary to obtain a small change $\Delta \eta$ of η . One gets

$$\Delta x_i = \Delta \eta / S_{x_i} \quad (17)$$

and hence the new value

$$x_{in} = x_{io} + \Delta x_i = x_{io} + (\Delta \eta / S_{x_i}). \quad (18)$$

The above formulas hold for systems which are linear or linearisable above the operating point. For highly nonlinear situations, the second order sensitivity coefficient

$$S'_{x_i} = dS_{x_i}/dx_i = d^2\eta/dx_i^2 \quad (19)$$

is necessary to preserve good accuracy. Use of S'_{x_i} together with S_{x_i} yields successively

$$\Delta \eta = S_{x_i} \Delta x_i + 0.5 S'_{x_i} (\Delta x_i)^2 \quad (20)$$

and

$$\Delta x_i = \frac{-S_{x_i} \pm \sqrt{(S_{x_i}^2 + 2S'_{x_i} \Delta \eta)}}{S'_{x_i}} \quad (21)$$

The above expressions may be used to assess the sensitivity with respect to any of the parameters on which η depends explicitly. According to eqn. 12 and Appendix 9.1, these explicit parameters are

$$P_m, \delta_0, \delta_\tau, \delta_P, \gamma, \ddot{\gamma}, v_0, v_D, v_P, P_{c0}, P_{cD}, P_{cP}, P_{max0}, P_{maxD}, P_{maxP} \quad (22)$$

On the other hand, in order to explore the sensitivity of η with respect to parameters depending implicitly on some of the above ones, we should replace eqns. 16 and 19, respectively, by

$$S_{x_i} = \frac{d\eta}{dx_i} = \frac{\partial\eta}{\partial x_i} + \sum \frac{\partial\eta}{\partial f(x_i)} \frac{df(x_i)}{dx_i} \quad (16')$$

$$S'_{x_i} = \frac{d^2\eta}{dx_i^2} = \frac{\partial^2 S_{x_i}}{\partial x_i^2} + \sum \frac{\partial S_{x_i}}{\partial f(x_i)} \frac{df(x_i)}{dx_i} \quad (19')$$

Discussion: Eqn. 21 calls for two remarks.

(a) The expression under the square root cannot assume negative values in the cases of concern, i.e. for real values of Δx_i .

(b) Of the two roots of eqn. 21, the one of concern is that with the smallest absolute value. This implies choice of the minus sign when S_{x_i} is negative.

3.2 Sensitivity of η with respect to the clearing time τ

For illustrative purposes let us consider this simple case with some detail. Since the margin η eqn. 12 depends implicitly on τ through δ_τ expressed by eqn. 13, we use eqn. 16' and obtain

$$S_\tau = \frac{d\eta}{d\tau} = \frac{\partial\eta}{\partial\delta_\tau} \frac{d\delta_\tau}{d\tau} = [P_{cD} - P_{cP} + P_{maxD} \sin(\delta_\tau - v_D) - P_{maxP} \sin(\delta_\tau - v_P)] \left[\alpha_1^{-1} \alpha_2^{-2} \gamma \tau + \frac{\alpha_1^{-1} \alpha_2^{-4} \ddot{\gamma} \tau^3}{6} \right] \quad (23)$$

Note that from a physical viewpoint S_τ should be negative in feasible situations. This is corroborated by the numerical results of Section 5.

The second-order sensitivity coefficient is derived by applying eqn. 19' to eqn. 23:

$$\begin{aligned} S'_\tau &= \frac{dS_\tau}{d\tau} = \frac{\partial S_\tau(\tau)}{\partial\tau} + \frac{\partial S_\tau(\tau)}{\partial\delta_\tau} \frac{d\delta_\tau}{d\tau} \\ &= [P_{cD} - P_{cP} + P_{maxD} \sin(\delta_\tau - v_D) - P_{maxP} \sin(\delta_\tau - v_P)] \left[\alpha_1^{-1} \alpha_2^{-2} \gamma + \frac{\alpha_1^{-1} \alpha_2^{-4} \ddot{\gamma} \tau^2}{2} \right] \\ &\quad + \left[\alpha_1^{-1} \alpha_2^{-2} \gamma \tau + \frac{\alpha_1^{-1} \alpha_2^{-4} \ddot{\gamma} \tau^3}{6} \right]^2 [P_{maxD} \cos(\delta_\tau - v_D) - P_{maxP} \cos(\delta_\tau - v_P)] \end{aligned} \quad (24)$$

3.3 Sensitivity of η with respect to the generation power P_{ms}

Let us first observe that the variations of the critical machine's generation P_{ms} are compensated by those of the remaining machines' generation $P_a = \sum_{l \in a} P_{ml}$. Applying eqn. 16' with $x_i = P_{ms}$ and $f(x_i) = P_a(P_{ms})$ yields

$$S_{P_{ms}} = \frac{\partial\eta}{\partial P_{ms}} + \frac{\partial\eta}{\partial P_a} \frac{dP_a}{dP_{ms}}$$

But by virtue of eqns. 28 and 29:

$$\frac{\partial\eta}{\partial P_a} = -\frac{M_s}{M_a} \frac{\partial\eta}{\partial P_{ms}} \quad \text{and} \quad \frac{dP_a}{dP_{ms}} = -1.$$

Hence

$$S_{P_{ms}} = \frac{M_T}{M_a} \frac{\partial\eta(P_m, \delta_P, \delta_0, \delta_\tau, \gamma, \ddot{\gamma})}{\partial P_{ms}} \quad (25)$$

where the parameters between brackets are all implicitly dependent on P_{ms} according to the expressions of Appendix 9.1. The various partial derivatives implied by eqn. 25 are sketched in Fig. 2. Their detailed derivation is given in Appendix 9.4.

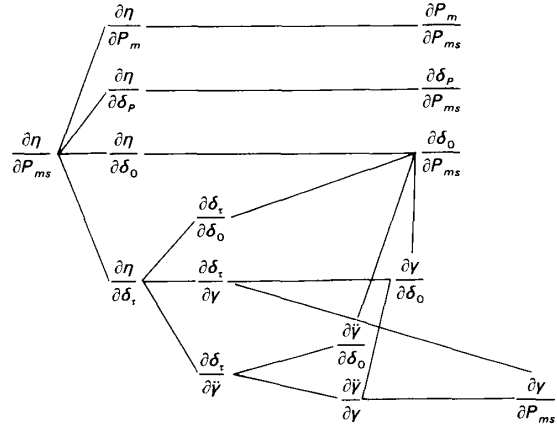


Fig. 2

Once again, physical reasoning suggests that $S_{P_{ms}}$ should be negative in feasible cases. Numerical results verify this conjecture.

In stringent, highly nonlinear situations, the second order sensitivity coefficient should also be considered. To make its derivation analytically tractable, we consider three-phase short-circuits located at those generators busbars which are found to compose the critical cluster, i.e. we consider the particular case where eqn. 39 holds. In this case, the derivation of the second order sensitivity coefficient

$$S'_{P_{ms}} = d^2\eta/dP_{ms}^2 \quad (26)$$

is expounded in Appendix 9.4. For other cases where eqn. 39 is not satisfied, the derived $S'_{P_{ms}}$ expression is only approximate. Nevertheless, its use proves to be beneficial in practice (see Section 5). The partial derivatives involved in eqn. 26 are sketched in Fig. 3.

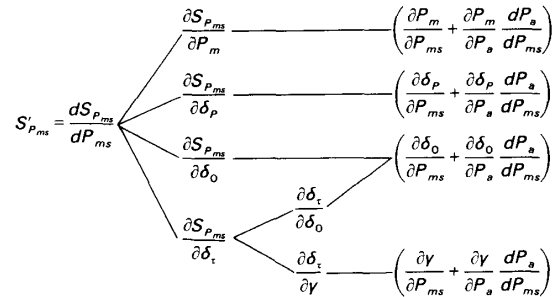


Fig. 3

Discussion: (a) The coefficient M_T/M_a appearing in eqn. 25 merely reflects the fact that the variations ΔP_{ms} are compensated by ΔP_a and shared among the ΔP_{ml} , $l \in A$.

(b) Since the multimachine system is replaced by a two-machine dynamic equivalent, the equivalent mechanical power of each group is equal to the sum of powers of the individual machines belonging to that group. The index explicitly depends on these two equivalent generations and so do the sensitivity coefficients. They are independent of the dispatch patterns within the same group.

(c) The above derivations hold both in the single and the multimachine critical cluster cases. But in this latter case one should treat the multimachine critical cluster in the same way as the aggregated group A : the constant parameters P_{max} , P_c , v and M_s are accordingly adapted and the P_{ms} variable represents a global value [3]; obviously, in such a case, the $S'_{P_{ms}}$ expression is only approximate.

3.4 Feasibility aspects of the sensitivity analysis

In the process of sensitivity coefficients derivation, various prerequisite conditions have been used. Let us summarise them and recall the methods by which they have been met.

(a) availability of a simple analytic method for transient stability assessment

(b) possibility of expressing analytically the stable and unstable equilibrium points of concern

(c) consideration of the sensitivity coefficients with respect to the generation of the reference busbar

(d) as far as possible, development of a non-iterative but accurate approach, despite the nonlinear character of the problem

These needs have been met as follows:

Problem (a) has been solved in References 3 and 4.

A method to meet problem (b) has been discussed in Section 2.3.

Problem (c) may be circumvented by observing that after the original load flow, the reference busbar concept is not used within the formulation of η , (exprs. 11 to 13) and of its derivatives.

To preserve accuracy despite the difficulties mentioned in (d), second order sensitivity coefficients have been considered as well as the first order ones.

4 Applications of the sensitivity analysis

A variety of interesting possible applications to both planning and online operation of power systems may be foreseen. Such uses of the sensitivity of η with respect to P_{ms} are tentatively suggested and discussed in this Section.

4.1 Generation limits

For a given operating condition, contingency and clearing time, the power generation limit P_{msl} may be computed by computing successively

(a) the corresponding η , via eqn. 12;

(b) the first and second order sensitivity coefficients $S_{P_{ms}}$, $S'_{P_{ms}}$, via eqns. 25 and 26;

(c) ΔP_{ms} corresponding to a chosen $\Delta\eta$ via eqn. 21;

(d) the power generation limit

$$P_{msl} = P_{mso} + \Delta P_{ms} \quad (27)$$

where P_{mso} denotes the value of P_{ms} at the operating point.

Generally, the correction ΔP_{ms} is sufficient to determine the generation limit P_{msl} . However, when the ratio $\Delta P_{ms}/P_{mso}$ takes on large values, it is advisable to proceed

to a new identification of the critical machine (or cluster) in order to check whether the new injection pattern leaves it unchanged. If not, a new procedure with the new critical machine has to be repeated.

For a list of contingencies and given clearing times the above procedure can be applied repeatedly, in order to draw up a chart of the power generation limits of the system. If a certain machine is found to be critical for several contingencies, the actual generation limit is the minimum among the limits corresponding to these contingencies.

4.2 Tie-line power flow limits

Based on the above generation limits and on appropriate distribution factors, one could determine the power flow limits on tie-lines. This important question is at present under investigation.

4.3 Transient security domain in the injection space

The generation limits of all n machines compose an n -dimensional vector. This vector defines a transient security domain in the injection space. The sum of its components provides the maximum load supply capability.

This allows one to reduce the online transient security assessment procedure to the steady-state security one: the system is guaranteed to be transiently secure, when all generations lie within the above defined generation limits.

4.4 Transient security dispatch

Security dispatch methods have been traditionally limited to steady-state contingency constraints [9]. Despite the attention focused on transient security dispatch, no realistic solution has as yet been proposed. Using the generation limits derived above, any existing or forthcoming steady-state security dispatch algorithm can be extended to transient security dispatch with negligible additional effort.

Indeed, for a given configuration, one merely should compute the above generation limit for each machine, compare it with the steady-state security limit and consider the smallest one to be the constraint of concern.

Remarks: (a) Economic dispatch with steady-state security constraints may thus readily be generalised into transient security domain by adding simple box constraints for individual generations. This is the easiest way to extend a steady-state security dispatch to a transient one; it can guarantee a designed transient security margin.

(b) Under a certain load pattern, the generation limits are independent of the dispatch among P_{ms} s; thus, all transient security constraints are fixed during the optimisation iterations.

4.5 The EEAC and derived sensitivity analysis as auxiliary tools

The aforementioned uses, along with that of EEAC, may be described as 'direct' applications of the proposed method. Their validity depends upon the validity of the simplified 'electromotive force behind transient reactance' model of the machines. Another family of applications, of the 'indirect' type, may also be considered. They concern the use of above tools as analytical means able to provide heuristics and attributes. These could be inserted within a more comprehensive artificial expert-like approach to a much more refined online transient stability analysis and

control. Such an application is considered in Reference 10.

5 Simulations

5.1 Test systems, simulation conditions, and accuracy assessment

Three different systems have been used so as to cover a variety of configurations, with real or realistic operating conditions. These are

(a) the CIGRE test system (7 machines, 10 busbars, 13 branches)

(b) the GREEK system (a modified version of the actual HV system, comprising 14 machines, 92 busbars, 119 branches)

(c) the IOWA reduced HV system (17 machines, 162 busbars, 281 branches).

The data for these systems are available upon request.

The disturbances considered are three-phase short-circuits (3 ϕ SC) located at either generator or non-generator busbars (labelled GB and OB respectively). Unless otherwise specified, the post-fault configuration is assumed to be identical to the pre-fault one.

The accuracy is assessed by using the time domain numerical integration method (labelled SBS for step-by-step) as the benchmark. To make sound comparisons, the same simplified machine model (electromotive force behind transient reactance) is used; the simulations are performed on the actual, unreduced, multimachine system. Three types of simulations are considered.

Type A: This type aims at assessing the goodness of the sensitivity coefficients $S_{P_{ms}}$, $S'_{P_{ms}}$, computed directly via the analytical expressions eqns. 25 and 26. To this end, the values thus found are compared with the values obtained by a purely numerical procedure. For example, to compute $S_{P_{ms}}$ one proceeds in a sequence as follows:

- fix $\Delta P_{ms} = +0.01$ p.u. (and also the ΔP_a as appropriate);
- compute the η corresponding to $P_{ms0} + \Delta P_{ms}$;
- compute $\Delta\eta = \eta - \eta_0$;
- compute $S_{P_{ms}} = \Delta\eta / \Delta P_{ms}$.

$S'_{P_{ms}}$ is computed numerically in a similar way.

Type B: Type B aims at assessing the goodness of the generation limit value, P_{msl} . This is achieved by considering an arbitrary value of τ and its corresponding margin η , then changing P_{ms} to P_{msl} so as to reduce the margin to zero, i.e. to make τ be the CCT of the system under the new operating condition corresponding to P_{msl} . The procedure comprises the following steps:

- fix a clearing time τ , generally different from the actual CCT;

- compute the corresponding margin η ;
- compute the sensitivity coefficients $S_{P_{ms}}$ about η , via the analytical expression eqn. 25;
- compute the corresponding ΔP_{ms} by using expr. 17 in which $\Delta\eta$ is chosen so as to reduce the new margin to zero;
- compute P_{msl} from eqn. 27;
- run a SBS for the system in the new operating condition imposed by P_{msl} to find the new CCT, labelled τ_1 ;
- the accuracy of P_{msl} is appraised by comparing τ_1 with τ .

Type C: This is the same as Type B except that $S'_{P_{ms}}$, is also considered, via eqn. 26, and the eqn. 21 instead of eqn. 17 is used to compute $\Delta\eta$.

Remark: In both Type A and Type C series of simulations, the second order sensitivity coefficients are computed even in the cases where the eqns. 26 and 42 are merely an approximation.

5.2 Simulation results

5.2.1 The CIGRE system: A first series of simulations are run by applying a 3 ϕ SC at successively all the system busbars, and by fixing the clearing time $\tau = 0.400$ s for all cases. This latter value has been chosen so as to get both stable and unstable cases ($\eta \neq 0$). The results reported in the Tables concern only those busbars for which the actual CCT is less than 1 s.

Table 1 summarises these results. It is divided into four parts. Part 1 describes the operating conditions, part 2 provides the comparisons of Type A simulations, part 3 those of Type B simulations, and part 4 those of Type C. Considering $S'_{P_{ms}}$ shows to improve the accuracy of GB and OB cases.

Table 2 deals specifically with all possible cases where the post-fault configuration is different from the pre-fault one, while preserving the system connectivity.

Another series of simulations concerns the exploration of the accuracy dependence on the clearing time. Two 3 ϕ SC are thus considered: one applied at GB #6 (initial CCT = 0.527), the other at OB #8 (initial CCT = 0.418). Table 3 collects the results obtained. The values of P_{msl} are also listed; the critical machine's busbar is explicitly indicated whenever it is different from the fault location, as for instance in the OB cases.

5.2.2 The GREEK system: Table 4 summarises results of simulations similar to those reported in Table 1. All GB cases are considered along with all OBs whose CCT does not exceed 1 s. As in Table 3, the values P_{msl} are given together with the critical busbar identification, whenever this busbar is different from the fault location.

Table 1: Accuracy assessment on the CIGRE system

1		2					3			4		
Fault at busbar number	type	CCT by SBS, (s)	$S_{P_{ms}}$ numer.	$S_{P_{ms}}$ anal.	$S_{P_{ms}}$ numer.	$S_{P_{ms}}$ anal.	P_{msl} $\times 100$ MW from eqn. 17	τ_1	$\frac{0.40 - \tau_1}{0.40}$ %	P_{msl} $\times 100$ MW from eqn. 21	τ_1	$\frac{0.40 - \tau_1}{0.40}$ %
1	GB	0.358	-11.0	-11.1	22.1	23.1	1.852	0.413	-3.2	1.918	0.399	0.2
2	GB	0.416	-16.2	-16.7	11.3	14.2	1.263	0.399	0.2	1.265	0.399	0.2
3	GB	0.395	-15.0	-15.3	12.5	13.4	2.517	0.402	-0.5	2.518	0.402	-0.5
4	GB	0.526	-19.2	-19.8	-5.4	-5.0	3.821	0.398	0.5	3.750	0.406	-1.5
5	GB	0.359	-11.1	-11.3	21.9	23.0	1.987	0.410	-2.5	2.049	0.399	0.2
6	GB	0.527	-15.8	-16.4	-4.2	-3.6	2.109	0.404	-1.0	2.084	0.408	-2.0
7	GB	0.339	-8.9	-9.0	11.0	12.0	1.426	0.408	-2.0	1.473	0.395	1.2
8	OB	0.481	-10.23	-10.92	1.89	-0.9	1.983	0.407	-1.8	1.981	0.405	-1.2

The clearing time τ has been fixed at 0.40 s for all busbars except for GB #84 for which $S_{P_{ms}}$ is found to be positive, which is physically unsound. This anomaly disappears for $\tau = 0.20$ s.

Another discrepancy is observed for the GB #1 (see in the last column the large percentage). In fact, the very

EEAC and the Liapunov criterion exhibit also inaccurate results for this busbar: CCT of respectively 0.49 s and 0.55 s whereas the actual CCT is 0.65 s. These very few cases where the EEAC method proves to be inaccurate are discussed in Reference 4. Note that the extensions newly proposed in reference 11 succeed in overcoming these difficulties.

Table 2: Accuracy assessment on the CIGRE system

fault at busbar number	line tripped in the post-fault configuration	P_{msl} $\times 100$ MW	τ_1 , s	$\frac{0.40 - \tau_1}{0.40}$ %
1	GB 1-3	1.852	0.398	0.5
	1-4	1.841	0.403	-0.7
2	GB 2-3	1.225	0.397	0.7
	2-10	1.256	0.401	-0.2
3	GB 3-1	2.473	0.395	1.2
	3-2	2.497	0.404	-1.0
	3-4	2.476	0.400	0.0
	3-9	2.510	0.400	0.0
4	GB 4-1	3.836	0.393	1.7
	4-3	3.843	0.394	1.5
	4-6	3.834	0.389	2.7
	4-9	3.849	0.393	1.7
	4-10	3.844	0.394	1.5
6	GB 6-4	1.960	0.385	3.7
	6-8	2.091	0.396	1.0
8	OB 8-6	1.861	0.399	0.2
	8-9	1.982	0.400	0.0

5.2.3 The IOWA system: On the Iowa system, 3 ϕ SCs have systematically been applied to all GBs and to some OBs randomly chosen. The clearing time has been uniformly fixed at $\tau = 0.30$ s, to give both stable and unstable cases.

Table 5 summarises the results. From these, the limit generation powers (rows relative to GBs, first column of part 3) are reproduced in Fig. 4, along with the corresponding P_{ms0} ; these are the values of P_{ms} which yield the CCTs quoted in the Table.

Fig. 4 thus gives an interesting overall picture of the system power injections and shows readily how far from instability (or stability) each generator is. Admittedly, the clearing time of 0.30 s used here may be deemed to be too large. A more sound picture should be drawn on the basis of clearing times fitting real world considerations.

The load supply capability of the system is obtained by summing up all the P_{msl} values of Fig. 4; it is found to be 18 446.3 MW, i.e. quite small. Again, this reflects the

Table 3: Accuracy assessment on the CIGRE system

fault at busbar number	type	τ s	1		2		3 P_{msl} $\times 100$ MW	τ_1 , s	$\frac{\tau - \tau_1}{\tau}$ %
			numer.	$S_{P_{ms}}$ anal.	numer.	$S'_{P_{ms}}$ anal.			
6	GB	0.1	-3.17	-3.27	-0.03	0.00	6.917	0.101	-1.0
	GB	0.2	-4.06	-4.65	-1.52	-1.37	4.283	0.201	-0.5
	GB	0.3	-8.31	-8.89	-4.81	-4.52	2.884	0.300	0.0
	GB	0.4	-15.84	-16.38	-4.29	-3.58	2.131	0.399	0.3
	GB	0.5	-20.58	-21.00	21.60	23.33	1.695	0.500	0.0
	GB	0.6	-8.16	-8.25	86.91	89.92	1.423	0.598	0.3
8	OB	0.1	-2.58	-3.06	0.34	0.23	3.992	0.102	-2.0
	OB	0.2	-3.54	-4.04	-1.40	-0.85	3.176	0.202	-1.0
	OB	0.3	-6.13	-6.09	-3.42	-2.85	2.467	0.301	-0.3
	OB	0.4	-10.27	-10.92	-1.91	-0.90	1.998	0.399	0.3
	OB	0.5	-12.14	-12.75	17.06	19.87	1.699	0.498	0.4
	OB	0.6	-3.88	-3.98	58.45	64.18	1.501	0.602	0.3

Table 4: Accuracy assessment on the GREEK system

		1		2				3		
fault at busbar number	type	CCT by SBS	τ , s	$S_{P_{ms}}$		$S'_{P_{ms}}$		P_{msl} $\times 100$ MW	τ_1 , s	$\frac{\tau - \tau_1}{\tau}$ %
		s	numer.	anal.	numer.	anal.				
1	GB	0.490	0.400	-1.46	-6.16	8.56	4.56	6.704 (G34)	0.343	14.2
33	GB	0.655	0.400	-12.00	-11.90	-12.89	-13.16	1.083	0.415	-3.7
34	GB	0.341	0.400	-7.78	-7.66	7.93	6.44	4.310	0.422	-5.5
64	GB	0.713	0.400	-10.20	-9.80	-13.01	-5.62	2.718	0.415	-3.7
76	GB	0.483	0.400	-12.24	-12.13	0.84	0.30	1.684	0.409	-2.2
17	GB	0.587	0.400	-10.38	-9.91	-114.00	-57.70	0.146	0.389	2.7
19	GB	0.527	0.400	-10.61	-9.99	-8.42	-11.25	0.486	0.386	3.5
21	GB	0.506	0.400	-12.34	-12.46	-2.09	-2.39	2.135	0.386	3.5
48	GB	>1.	0.400	-8.23	-7.57	-178.21	-200.85	0.079	0.412	-3.0
59	GB	0.488	0.400	-13.08	-13.12	-0.64	-1.47	0.978	0.395	1.2
77	GB	0.487	0.400	-10.32	-10.02	-0.65	-1.51	0.553	0.401	-0.2
84	GB	0.268	0.200	-5.28	-5.36	-1.68	-0.14	5.440	0.203	-1.5
89	GB	0.308	0.400	-4.56	-4.61	10.30	9.93	1.514	0.417	-4.2
0	GB	0.750	0.400	-8.73	8.21	-4.90	-3.34	4.989	0.383	4.2
72	OB	0.375	0.400	-9.42	-9.61	5.57	5.08	5.139 (G34)	0.428	-7.0
91	OB	0.413	0.400	-8.93	-8.41	2.63	3.43	5.838 (G34)	0.393	1.7
71	OB	0.386	0.400	-10.08	-9.73	4.96	4.71	5.273 (G34)	0.426	-6.5
37	OB	0.356	0.400	-9.04	-8.84	7.15	5.85	4.815 (G34)	0.422	-5.5
22	OB	0.774	0.400	-8.80	-8.82	-6.71	-7.21	2.973 (G34)	0.396	1.0

Table 5: Accuracy assessment on the IOWA system

1		2					3		
fault at busbar name	CCT by SBS, type s	$S_{P_{ms}}$ numer.	$S_{P_{ms}}$ anal.	$S'_{P_{ms}}$ numer.	$S'_{P_{ms}}$ anal.	$P_{ms/100}$ MW critical machine	τ_1 ,	$\frac{0.30 - \tau_1}{0.30}$ %	
G1	↑	0.29	-8.33	-9.26	0.02	0.02	21.808	0.28	6.7
G2	•	0.22	-5.31	-5.87	0.80	1.20	6.342	0.29	3.3
G3	•	0.29	-6.54	-6.89	0.18	0.16	14.378	0.30	0.0
G4	•	0.40	-11.13	-11.09	-0.11	-0.17	19.176	0.31	-3.3
G5	•	0.24	-6.26	-6.78	2.80	3.54	3.556	0.30	0.0
G6	•	0.21	-4.40	-4.65	1.80	2.26	7.646	0.29	3.3
G7	•	0.31	-8.76	-9.55	1.34	1.24	1.396	0.29	3.3
G8	•	0.23	-7.16	-7.26	18.72	23.51	0.639	0.28	6.7
G9	GB	0.26	-5.71	-6.27	0.40	0.55	4.993	0.29	3.3
G10	•	0.25	-6.71	-7.57	9.11	11.85	1.075	0.30	0.0
G11	•	0.35	-7.01	-7.75	-1.84	-0.32	2.058	0.30	0.0
G12	•	0.24	-5.75	-6.19	2.94	3.72	4.757	0.30	0.0
G13	•	0.40	-6.32	-6.50	-0.06	-0.09	33.472	0.29	3.3
G14	•	0.41	-5.85	-6.14	-0.06	-0.08	30.844	0.29	3.3
G15	•	0.28	-9.30	-9.39	0.33	0.27	23.192	0.30	0.0
G16	•	0.32	-6.10	-6.78	0.22	0.31	4.755	0.31	-3.3
G17	↓	0.23	-5.34	-5.78	2.48	3.28	4.396	0.30	0.0
COOPER	↑	0.23	-5.41	-6.00	0.73	1.09	6.556 (G2)	0.29	3.3
NEBRCY	•	0.23	-5.60	-6.07	2.36	3.15	4.485 (G17)	0.30	0.0
RAUN	•	0.21	-4.61	-4.90	1.79	2.22	7.772 (G6)	0.28	6.7
CALHOUN	•	0.35	-5.98	-6.67	0.08	0.13	4.991 (G16)	0.32	-6.7
112	•	0.24	-6.20	-6.70	2.79	3.55	4.919 (G12)	0.30	0.0
52	•	0.36	-6.95	-7.63	-0.41	-0.59	2.115 (G11)	0.30	0.0
ARNOLD	•	0.26	-5.68	-6.26	0.38	0.51	5.052 (G9)	0.29	3.3
95	•	0.32	-8.73	-9.53	1.12	1.39	1.418 (G7)	0.29	3.3
96	OB	0.23	-6.71	-7.31	18.58	23.35	0.642 (G8)	0.28	6.7
NEAL	•	0.24	-6.45	-7.00	2.67	3.39	3.634 (G5)	0.30	0.0
42	•	0.36	-9.92	-4.40	0.11	-0.12	6.327 (G9)	0.30	0.0
110	•	0.25	-7.31	-8.04	8.06	10.57	1.128 (G10)	0.31	-3.3
120	•	0.27	-6.79	-7.43	1.24	2.81	5.498 (G12)	0.30	0.0
128	•	0.33	-5.78	-6.38	0.68	1.77	6.485 (G12)	0.32	-6.7
149	•	0.25	-6.50	-7.06	2.60	3.33	3.661 (G5)	0.30	0.0
98	•	0.44	-3.28	-3.73	0.15	-0.23	6.249 (G9)	0.33	-10.0
150	↓	0.42	-4.81	-5.40	0.31	0.71	5.307 (G5)	0.33	-10.0

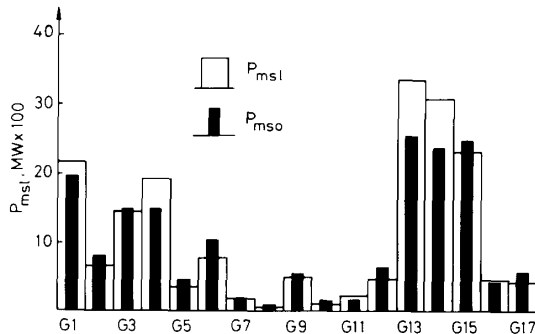


Fig. 4 Limit (P_{msl}) and actual (P_{mso}) generation power injections in the IOWA system

□ P_{msl}
■ P_{mso}

large value of τ . For example, let us indicate that for $\tau = 0.12$ s the load supply capability is found to be 33 689.4 MW.

Repeating the same computation of P_{msl} for various values of τ , we also obtain the curves P_{msl} against τ , $s = 1, 2, \dots, n$, and $\sum_{s=1}^n P_{msl}$ against τ . Figs. 5a and b, respectively, show these two types of curves. These pictures are very helpful for planners to know how to improve the protection relay system, as well as for operators to know the margin of load supply capability. Note that a similar curve, providing the load supply capability of the GREEK system has proven to be extremely useful

for the automatic generation of a first learning set, within an artificial intelligence-like approach [10].

5.2.4 Discussion: The above results suggest the following observations.

(a) Coefficients $S_{P_{ms}}$ are quite accurate for both GBs and OBs.

(b) $S'_{P_{ms}}$ is generally less accurate than $S_{P_{ms}}$, especially for OBs; this is normal since the expression of $S'_{P_{ms}}$ is rigorous only for a subclass of the GBs (see Section 3.3).

(c) The closer the τ to the actual CCT, the more accurate the results. This is obviously due to the nonlinear shape of the curve η against τ .

(d) Overall, the results are very satisfactory.

5.3 Sensitivity coefficients S_τ , S'_τ , and related CCT estimations

Table 6 gives the results of S_τ and S'_τ and related CCT estimations for the CIGRE system, where t_{c1} and t_{c2} , respectively, stand for the CCT estimation on the basis of S_τ alone, and of S_τ together with S'_τ . Obviously, consideration of S'_τ is necessary to get good accuracy.

The results obtained for $\tau = 0.20$ s are not as accurate as those relative to $\tau = 0.40$ s: apparently the slope of η against τ is strongly nonlinear and $\tau = 0.20$ is too far from CCT to obtain it in a single step. Let us remark that the purpose of S_τ and S'_τ is not to compute CCTs (which is readily obtained through the procedure of Reference 4) but rather to appraise their variation with the clearing time. For this purpose, the sensitivity calculations are interesting, although not as accurate as those relative to P_{ms} .

Table 6: Accuracy assessment on the CIGRE system

fault number	at busbar types	CCT by SBS	S_t	S'_t	$\tau = 0.2$ s		S_t	S'_t	$\tau = 0.4$ s	
					t_{c1}	t_{c2}			t_{c1}	t_{c2}
1	GB	0.358	-45	-512	0.490	0.355	-65	1208	0.345	0.360
2	GB	0.416	-19	-223	0.727	0.426	-61	113	0.417	0.418
3	GB	0.394	-40	-352	0.617	0.414	-100	231	0.394	0.394
4	GB	0.526	-18	-158	1.681	0.678	-93	-310	0.575	0.541
5	GB	0.359	-45	-511	0.494	0.356	-66	1192	0.347	0.361
6	GB	0.527	-12	-96	1.428	0.646	-58	-179	0.541	0.519
7	GB	0.339	-21	-215	0.443	0.342	-41	347	0.331	0.344
8	OB	0.418	-7	-77	1.070	0.519	-29	-94	0.489	0.479

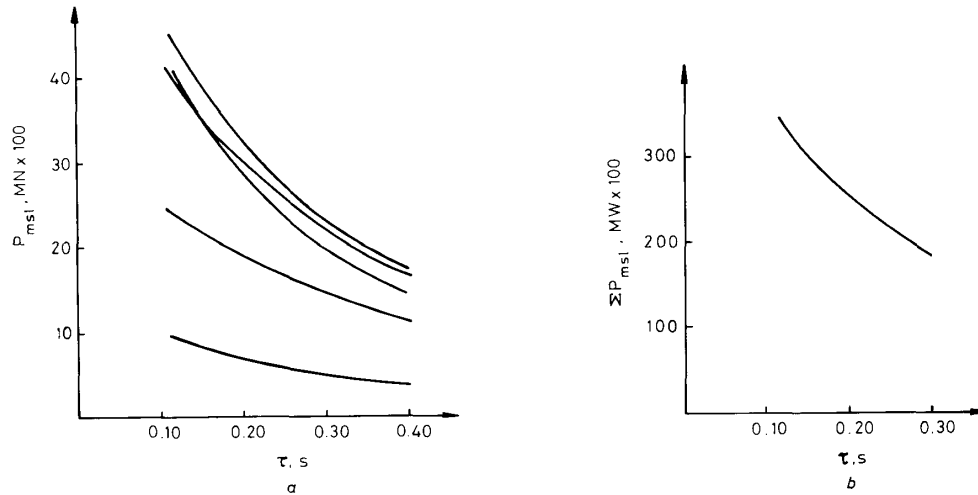


Fig. 5 (a) Generation limits of individual machines of the Iowa system (b) Load supply capability of the Iowa system

5.4 Estimation of CPU computing times

The IOWA system, which is one of our largest test systems, is chosen to estimate computing times. The computer used is a DEC-20.

The time required for the computation of $S_{P_{ms}}$, $S'_{P_{ms}}$, S_t , S'_t , and P_{msl} altogether is about 0.02 s per contingency; it is found to be almost independent of the system size. This attractive figure is due to the simple algebraic formulation of the sensitivity coefficients. Notice incidentally that the time required for the computation of η for 30 contingencies is 11.63 s. This time is found to vary almost linearly with the system size.

Let us finally recall the conclusion reported in Reference 4, relative to the computation of CCTs by the EEAC. It is found that when the pre-fault and post-fault configurations are identical, the gain in computing time of the EEAC over the most efficient Liapunov criterion is about one order of magnitude for running 30 contingencies. The gain is even more important when the post-fault configuration is different from the pre-fault one.

6 Conclusions

An analytic and straightforward sensitivity method has been proposed. It has been shown to be extremely fast and to provide reliable, sound, and abundant information about the transient robustness of a power system. Among its many possible uses, let us quote the following:

the ordering of the machines most affected by the disturbance, and the margin corresponding to each individual machine;

the margin for the global system;

the first and second order sensitivity coefficients of the margin with respect to clearing time;

the first and second order sensitivity coefficients of the margin with respect to individual active power generation;

the generation limit of the critical machine;

the additional generation allowed or the drop of generation needed to keep the margin at a desired value.

Moreover, applying the method to a set of selected contingencies, enables one to reach the following objectives:

(a) Obtain the maximum load supply capability and define the exact (necessary and sufficient) transient security region in the injection space. In this case, the online transient stability monitoring simply reduces to check whether the present injection vector lies within this region, likewise in steady-state security monitoring.

(b) Extend any steady-state security dispatch algorithm to transient security dispatch, by combining generation limits dictated by both of the above security aspects.

All these contributions can greatly help planners and operators to make decisions within a time frame convenient to online requirements.

In addition to the above 'direct' applications, the method can also provide auxiliary tools to be inserted within a more general and refined approach to transient stability assessment and control.

This latter aspect, as well as further exploration of the sensitivity method, is at present under investigation.

7 Acknowledgment

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9 Appendix: Matters relating to the EEAC

9.1 Symbols and formulas

The symbols of the equivalent one-machine-infinite-busbar system considered in Section 2 are

$$M = M_s M_a M_T^{-1}, \quad M_T = \sum_{i=1}^n M_i \quad (28)$$

$$\left. \begin{aligned} P_m &= \left(M_a P_{ms} - M_s \sum_{l \in A} P_{ml} \right) M_T^{-1} \\ P_c &= \left[M_a E_s^2 G_{ss} - M_s \left(\sum_{l \in A} E_l^2 G_{ll} + \sum_{l, j \in A; j \neq l} E_l E_j G_{lj} \right) \right] M_T^{-1} \\ P_{max} &= \left[\sum_{l \in A} E_s E_l Y_{sl} (M_s^2 + M_a^2 - 2 M_s M_a \cos 2\theta_{sl})^{1/2} \right] M_T^{-1} \\ v &= \tan^{-1} \left(\frac{M_T}{M_a - M_s} \tan \theta_{s,a} \right) - \frac{\pi}{2}, \\ \theta_{s,a} &= \text{the argument of the equivalent admittance between } s \text{ and } a \\ \tan \theta_{s,a} &= \left(\sum_{l \in A} B_{sl} \right) \left(\sum_{l \in A} G_{sl} \right)^{-1}, \end{aligned} \right\} \quad (29)$$

where B stands for susceptance, G for conductance:

$$G_{sl} = Y_{sl} \cos \theta_{sl}; \quad B_{sl} = Y_{sl} \sin \theta_{sl}. \quad (30)$$

The accelerating area, A_{acc} , and decelerating area, A_{dec} , are expressed by

$$A_{acc} = (P_m - P_{cD})(\delta_\tau - \delta_0) + P_{maxD}[\cos(\delta_\tau - v_D) - \cos(\delta_0 - v_D)] \quad (31)$$

$$A_{dec} = (P_{cP} - P_m)(\pi - \delta_\tau - \delta_P + 2v_P) + P_{maxP}[\cos(\delta_\tau - v_P) + \cos(\delta_P - v_P)] \quad (32)$$

Eqn. 13 relating δ_τ to τ involves the following quantities:

$$\begin{aligned} (a) \quad \gamma &\triangleq \gamma \Big|_{t=0^+} = \frac{d^2 \delta}{dt^2} \Big|_{t=0^+} = M_s^{-1} \left[P_{ms} - E_s^2 Y_{ssD} \cos \theta_{ssD} - E_s \cos(\delta_0 - \theta_D) \sum_{l \in A} E_l Y_{slD} \right] \\ &\quad - M_a^{-1} \left[\sum_{l \in A} P_{ml} - \sum_{l \in A} \sum_{j \in A} E_l E_j Y_{ljD} \cos \theta_{ljD} - E_s \cos(\delta_0 + \theta_D) \sum_{l \in A} E_l Y_{slD} \right] \end{aligned} \quad (33)$$

$$\begin{aligned} (b) \quad \ddot{\gamma} &\triangleq \frac{d^2 \gamma}{dt^2} \Big|_{t=0^+} = [M_s^{-1} \sin(\delta_0 - \theta_D) - M_a^{-1} \sin(\delta_0 + \theta_D)] E_s \gamma \sum_{l \in A} E_l Y_{slD} \end{aligned} \quad (34)$$

where 0^+ stands for the time following immediately the disturbance inception and $\delta_{0^+} = \delta_{0^-} = \delta_0$;

(c) the angle factor α_1 defined by

$$\alpha_1 = (\delta_x - \delta_0)/(\delta_\tau - \delta_0) \quad (35)$$

with $\delta_0 < \delta_x < \delta_\tau$;

obviously, $0 < \alpha_1 < 1$;

(d) the time factor $\alpha_2 (\alpha_2 > 1)$ defined by

$$\alpha_2 = \tau/t_x. \quad (36)$$

These factors are introduced to counterbalance truncation effects. Their values are related by $\alpha_1 \alpha_2^2 = 1$; they can be fixed once for all, as suggested in Reference 4.

9.2 Computation of the critical clearing time (CCT)

For a given disturbance and the corresponding critical cluster, the CCT computation amounts to (a) computing the critical clearing angle, δ_c , by setting $\eta = 0$ and solving eqn. 12 with $\delta_\tau \equiv \delta_c$, and (b) computing $t_c \equiv$ CCT by setting in eqn. 13 $\tau = t_c$ and solving it with respect to t_c .

This computation is extremely fast; the required time is almost totally negligible compared to that for the Y matrices' reduction; this reduction amounts to eliminating all busbars except those internal to the generator nodes.

9.3 Identification of the critical cluster

For a given disturbance this identification is carried out as follows:

(a) draw up a list of 'candidate' critical clusters, by using the initial accelerations (eqns. 33) to 'pre-filter' the plausible candidates;

(b) compute by turn the corresponding candidate critical clearing times: the actual critical cluster will be the one yielding the smallest of the above values.

Thus, the identification of the critical cluster amounts to using the procedure of Appendix 9.2 repeatedly; the required computing effort is still almost negligible.

9.4 Derivation of sensitivity coefficients

In view of Fig. 2, the partial derivative $\partial\eta/\partial P_{ms}$ in eqn. 25 takes the form

$$\frac{\partial\eta}{\partial P_{ms}} = \frac{\partial\eta}{\partial P_m} \frac{\partial P_m}{\partial P_{ms}} + \frac{\partial\eta}{\partial \delta_P} \frac{\partial \delta_P}{\partial P_{ms}} + \left[\frac{\partial\eta}{\partial \delta_0} + \frac{\partial\eta}{\partial \delta_\tau} \left(\frac{\partial \delta_\tau}{\partial \delta_0} + \frac{\partial \delta_\tau}{\partial \gamma} \frac{\partial \gamma}{\partial \delta_0} + \frac{\partial \delta_\tau}{\partial \ddot{\gamma}} \left(\frac{\partial \ddot{\gamma}}{\partial \delta_0} + \frac{\partial \ddot{\gamma}}{\partial \gamma} \frac{\partial \gamma}{\partial \delta_0} \right) \right) \right] \frac{\partial \delta_0}{\partial P_{ms}} + \frac{\partial\eta}{\partial \delta_\tau} \left[\frac{\partial \delta_\tau}{\partial \gamma} + \frac{\partial \delta_\tau}{\partial \ddot{\gamma}} \frac{\partial \ddot{\gamma}}{\partial \gamma} \frac{\partial \gamma}{\partial P_{ms}} \right] \quad (37)$$

where the various partial derivatives are explicated as follows:

$$\left. \begin{aligned} \frac{\partial\eta}{\partial P_m} &= \delta_0 + \delta_P - \pi - 2v_P; \quad \frac{\partial\eta}{\partial \delta_P} = P_m - P_{cP} - P_{maxP} \sin(\delta_P - v_P) \quad \frac{\partial\eta}{\partial \delta_0} = P_m - P_{cD} - P_{maxD} \sin(\delta_0 - v_D) \\ \frac{\partial\eta}{\partial \delta_\tau} &= P_{cD} - P_{cP} + P_{maxD} \sin(\delta_\tau - v_D) - P_{maxP} \sin(\delta_\tau - v_P) \quad \frac{\partial \delta_\tau}{\partial \delta_0} = 1; \quad \frac{\partial \delta_\tau}{\partial \gamma} = \frac{\tau^2}{2\alpha_1\alpha_2^2}; \\ \frac{\partial \delta_\tau}{\partial \ddot{\gamma}} &= \frac{\tau^4}{24\alpha_1\alpha_2^2} \frac{\partial \ddot{\gamma}}{\partial \delta_0} = [M_s^{-1} \cos(\delta_0 - \theta_D) - M_a^{-1} \cos(\delta_0 + \theta_D)] \gamma E_s \sum_{l \in A} E_l Y_{slD} \\ \frac{\partial \ddot{\gamma}}{\partial \gamma} &= [M_s^{-1} \sin(\delta_0 - \theta_D) - M_a^{-1} \sin(\delta_0 + \theta_D)] E_s \sum_{l \in A} E_l Y_{slD} \\ \frac{\partial \gamma}{\partial \delta_0} &= \frac{\partial \ddot{\gamma}}{\partial \gamma}; \quad \frac{\partial P_m}{\partial P_{ms}} = \frac{M_a}{M_T}; \quad \frac{\partial \delta_P}{\partial P_{ms}} = \frac{M_a}{M_T P_{maxP} \cos(\delta_P - v_P)} \\ \frac{\partial \delta_0}{\partial P_{ms}} &= \frac{M_a}{M_T P_{max0} \cos(\delta_0 - v_0)}; \quad \frac{\partial \gamma}{\partial P_{ms}} = \frac{1}{M_s} \end{aligned} \right\} \quad (38)$$

In the case where

$$P_{maxD} = 0 \quad (39)$$

the first order sensitivity coefficient expressed by eqns. 25 and 37 reduces to

$$\begin{aligned} S_{P_{ms}} = \frac{d\eta}{dP_{ms}} &= \delta_0 + \delta_P - \pi - 2v_P + \frac{P_m - P_{cP}}{P_{maxP} \cos(\delta_P - v_P)} - tg(\delta_P - v_P) \\ &+ \frac{P_m - P_{cP} - P_{maxP} \sin(\delta_\tau - v_P)}{P_{max0} \cos(\delta_0 - v_0)} + [P_{cD} - P_{cP} - P_{maxP} \sin(\delta_\tau - v_P)] \frac{\tau^2 M_T}{2\alpha_1\alpha_2^2 M_a M_s} \end{aligned} \quad (40)$$

Hence, the second order sensitivity coefficient is expressed by (see Fig. 3):

$$\begin{aligned} S'_{P_{ms}} = \frac{d^2\eta}{dP_{ms}^2} &= \frac{dS_{P_{ms}}}{dP_{ms}} = \frac{\partial S_{P_{ms}}}{\partial P_m} \left(\frac{\partial P_m}{\partial P_{ms}} + \frac{\partial P_m}{\partial P_a} \frac{dP_a}{dP_{ms}} \right) + \frac{\partial S_{P_{ms}}}{\partial \delta_P} \left(\frac{\partial \delta_P}{\partial P_{ms}} + \frac{\partial \delta_P}{\partial P_a} \frac{dP_a}{dP_{ms}} \right) \\ &+ \left(\frac{\partial S_{P_{ms}}}{\partial \delta_0} + \frac{\partial S_{P_{ms}}}{\partial \delta_\tau} \frac{\partial \delta_\tau}{\partial \delta_0} \right) \left(\frac{\partial \delta_0}{\partial P_{ms}} + \frac{\partial \delta_0}{\partial P_a} \frac{dP_a}{dP_{ms}} \right) + \frac{\partial S_{P_{ms}}}{\partial \delta_\tau} \frac{\partial \delta_\tau}{\partial \gamma} \left(\frac{\partial \gamma}{\partial P_{ms}} + \frac{\partial \gamma}{\partial P_a} \frac{dP_a}{dP_{ms}} \right). \end{aligned} \quad (41)$$

The various partial derivatives are given below.

$$\left. \begin{aligned} \frac{\partial S_{P_{ms}}}{\partial P_m} &= \frac{1}{P_{maxP} \cos(\delta_P - v_P)} + \frac{1}{P_{max0} \cos(\delta_0 - v_0)} \\ \frac{\partial S_{P_{ms}}}{\partial \delta_P} &= 1 + \frac{(P_m - P_{cP}) \sin(\delta_P - v_P)}{P_{maxP} \cos^2(\delta_P - v_P)} - \frac{1}{\cos^2(\delta_P - v_P)} \\ \frac{\partial S_{P_{ms}}}{\partial \delta_0} &= 1 + \frac{[P_m - P_{cP} - P_{maxP} \sin(\delta_\tau - v_P)] \sin(\delta_0 - v_0)}{P_{max0} \cos^2(\delta_0 - v_0)} \\ \frac{\partial S_{P_{ms}}}{\partial \delta_\tau} &= - \left(\frac{1}{P_{max0} \cos(\delta_P - v_P)} - \frac{\tau^2 M_T}{2\alpha_1\alpha_2^2 M_s M_a} \right) P_{maxP} \cos(\delta_\tau - v_P) \\ \frac{\partial P_m}{\partial P_a} &= - \frac{M_s}{M_T}; \quad \frac{dP_a}{dP_{ms}} = -1; \quad \frac{\partial P_m}{\partial P_{ms}} + \frac{\partial P_m}{\partial P_a} \frac{dP_a}{dP_{ms}} = 1 \\ \frac{\partial \delta_P}{\partial P_{ms}} + \frac{\partial \delta_P}{\partial P_a} \frac{dP_a}{dP_{ms}} &= \frac{1}{P_{maxP} \cos(\delta_P - v_P)} \\ \frac{\partial \delta_0}{\partial P_{ms}} + \frac{\partial \delta_0}{\partial P_a} \frac{dP_a}{dP_{ms}} &= \frac{1}{P_{max0} \cos(\delta_0 - v_0)} \\ \frac{\partial \gamma}{\partial P_{ms}} + \frac{\partial \gamma}{\partial P_a} \frac{dP_a}{dP_{ms}} &= \frac{M_T}{M_a M_s} \end{aligned} \right\} \quad (42)$$